Supplementary Material

Relevant, But Not Useful: Prospective Utility in Agentic Search

Paper ID: 507 (Anonymous submission)

This supplement provides implementation details, the preregistration and provenance protocol, the agent compliance audit, exact feature definitions, additional figures, and secondary results referenced in the main paper. Section 1 details the agent and retrieval stack; Section 2 the candidate families; Section 3 the preregistration chain and lockbox protocol; Section 4 the cross-family compliance audit; Section 5 pairwise label construction; Section 6 feature definitions; Section 7 the exact-value tables referenced from the main paper (headroom and signature coefficients); Section 8 the full trajectory of the case study; Section 9 additional figures; Section 10 additional quantitative results; and Section 11 reproducibility information.

1 Agent and Retrieval Stack

Search agent. The agent follows a ReAct-style loop: at each turn it emits either a search action (a query wrapped in `<search>` tags) or a final answer (wrapped in `<answer>` tags). Each search returns the top-3 documents (title and abstract text) from the fixed corpus, appended to the context as an observation block. Episodes allow at most 4 search turns; decoding is greedy (temperature 0) with stop sequences on the action delimiters and on generation of an observation-opening tag, so that the model cannot hallucinate its own retrieval results. Unclosed action tags at end-of-generation are parsed conservatively (an unclosed `<search>`/`<answer>` span is accepted only when it is unambiguous; occurrences are counted and audited). A same-code control comparison between parser versions showed no evidence of a behavioral effect of these parsing safeguards, and no safeguard-specific mechanism activated in the confirmatory runs.

Corpus. All settings use the official HotpotQA `enwiki-20171001` abstract corpus (5,233,235 documents). 2WikiMultiHopQA and MuSiQue are evaluated against the same fixed corpus; annotated supporting documents are covered for 92.7% of 2WikiMultiHopQA and 98.0% of MuSiQue validation questions, and confirmatory 2WikiMultiHopQA evaluation is restricted to covered questions (main paper, Sec. 4.1).

Retrievers. Lexical retrieval uses BM25 (`bm25s`). Dense retrieval uses `intfloat/e5-base-v2` with brute-force inner product over the full corpus (fp32). The relevance baseline is the MS MARCO cross-encoder `cross-encoder/ms-marco-MiniLM-L6-v2`, scoring each candidate's title and text against the current query q_t. All retriever configurations are fixed across the five evaluation settings.

2 Candidate Families

For every eligible state, the retrieval pool $\mathcal{C}_t$ is the union of BM25 top-20 lists under q_t, q_0, and up to six already-seen document titles (entity-style queries), plus dense top-20 lists under q_t and q_0, excluding all previously consumed documents (109.1 ± 27.7 documents per state in the primary evaluation). Up to eight typed candidates are drawn per state:

- **bm25_top** — BM25 top-1 under q_t;
- **dense_top** — dense top-1 under q_t;
- **ce_top** — cross-encoder argmax over $\mathcal{C}_t$ (the relevance-selection baseline);
- **mid_rel** — a BM25 rank 9–16 document under q_t;
- **novelty** — in the pool but outside BM25 top-20 under q_t and outside dense top-5 (low current-query relevance, reached via q_0/dense-tail expansion);

- **redundancy** — high q_t relevance (BM25 top-10) with the highest lexical (Jaccard) overlap against already-seen evidence (retained only if overlap > 0.05);
- **entity_bridge** — reached only via entity-title queries and absent from the BM25 ranking under q_t;
- **random** — a random remaining pool member (control).

Consistency of the CE baseline. The **ce_top** family is constructed so that the CE argmax over the full pool $\mathcal{C}_t$ is always contained in the retained intervention set $\mathcal{D}_t$: candidates are drawn without replacement, and if the pool-level CE argmax has already been taken as **bm25_top** or **dense_top**, it remains in $\mathcal{D}_t$ under that label while **ce_top** takes the next-ranked document. Consequently, $\arg\max_{d \in \mathcal{D}_t} R_{\mathrm{CE}}(q_t, d)$ and $\arg\max_{d \in \mathcal{C}_t} R_{\mathrm{CE}}(q_t, d)$ select the same document, and the main paper's d_{CE} (defined over $\mathcal{D}_t$) equals the pool-level CE selection. We verified this empirically: in all 3,881 evaluated states across the five settings, the pool-level CE argmax is present in $\mathcal{D}_t$ (zero exceptions).

Standard-retrieval-only analyses restrict the oracle to {bm25_top, dense_top, ce_top}; leave-one-family-out analyses remove one family at a time (main paper, Sec. 5.5).

3 Preregistration, Lockbox, and Provenance

Preregistration chain. All confirmatory decisions were fixed in a version-controlled design-revisions document *before* the corresponding data were inspected, including: the preference threshold $\tau = 0.1$ with sensitivity values fixed in advance; the primary Gate-2 endpoints; the lockbox endpoints (H1–H3) and the analysis freeze list; the predictor judgment bands and the pre-committed decision to stop the predictor line if strict criteria were not met (they were not; the result is reported as a negative finding rather than retried); the replication protocols for 2WikiMultiHopQA, the cross-scale agent, and MuSiQue; and the final review-defense analyses (cross-fitted oracle, question-clustered inference, pool-robustness, and effect-size decomposition). No acceptance criterion was relaxed after unblinding at any point; criteria that failed remain recorded as failures together with the adjudication that followed.

Single-open lockbox. The held-out HotpotQA lockbox was sealed before predictor development: question IDs were selected and committed, all models to be evaluated were frozen to disk, and SHA-256 hashes of the frozen artifacts and the design document were recorded in a provenance file within the repository before opening. The evaluation script refuses to run if a lockbox report already exists, so the lockbox was evaluated exactly once.

Measurement-integrity control. A preregistered harness audit compared agent behavior across parser versions. The strict no-op criterion (99% identical trajectories) was *not* met (94%); a preregistered same-code control (re-running the identical legacy code twice under production conditions) yielded 76% identical trajectories, attributing the discrepancy to serving-level nondeterminism, and zero activations of the new parsing mechanisms were observed in confirmatory runs. We therefore report "no evidence of a behavioral effect" rather than a proven no-op, and record the original criterion as failed.

4 Cross-Family Compliance Audit

Cross-family agent replications were subject to preregistered tool-protocol compliance criteria evaluated on a 30-question smoke set before any confirmatory run (malformed action rate $\leq 5\%$, valid answer rate, token-limit truncation rate, and zero answer leakage). Table 1 summarizes the audit. Only Qwen2.5-14B passed; Mistral-7B-v0.3 and Llama-3.1-8B were excluded *before* confirmatory evaluation, and the preregistration prohibits re-measuring an excluded model. Consequently, the main paper claims cross-scale transfer within the Qwen2.5 family only.

5 Pairwise Label Construction

For candidates d_i, d_j at the same state, the preference label is decided by the controlled utility difference with a preregistered tie zone $\tau = 0.1$: pairs with $|U_i - U_j| \leq \tau$ are ties and are excluded from accuracy denominators. Label reliability on the replay-audited subset is 97.8–99.0% (three greedy continuations per

Table 1: Compliance audit for candidate cross-family agents (30-question preregistered smoke evaluation; criteria fixed in advance).

Agent	Malformed	Valid answer	Token-limit	Decision
Qwen2.5-7B-Instruct	(reference)	—	—	primary agent
Mistral-7B-Instruct-v0.3	42.2%	100%	0%	excluded
Llama-3.1-8B-Instruct	5.9%	96.7%	3.3%	excluded (near boundary)
Qwen2.5-14B-Instruct	0.0%	93.3%	6.7%	**passed**

intervention on 20% of states; 115/579 primary states and 137/686 lockbox states). The distribution of preference strengths is bimodal (most decided pairs are either near-ties or large-margin), and threshold sensitivity at $\tau \in \{0.05, 0.2\}$ preserves the direction of all decided comparisons. Pair representations used by learned models are constructed to be orientation-invariant (candidates serialized in canonical order; magnitude-based pair features), which removes label-orientation leakage; an early signed-difference variant was discarded for this reason before any confirmatory use.

6 State-Delta Feature Definitions

The interpretable feature vector $\varphi(s_t, d)$ asks what the candidate *adds* relative to current knowledge. Content tokens are lowercased alphanumeric tokens with a fixed stopword list removed; entity proxies are capitalized tokens and multi-digit numbers. With D the candidate's content-token set, $E_1, \ldots, E_k$ the per-document evidence token sets ($E = \bigcup_i E_i$), Q and Q_t the token sets of the original question and current query:

- **novelty** $= |D \setminus E|/|D|$ — fraction of candidate content unseen in evidence;
- **redundancy** $= \max_i \text{Jaccard}(D, E_i)$ — maximal overlap with any single consumed document;
- **entity_gain** — new entity-proxy types per 100 tokens (scale-clipped);
- **query_align** $= |D \cap Q_t|/|Q_t|$;
- **q0_align** $= |D \cap Q|/|Q|$;
- **gap_coverage** $= |D \cap (Q \setminus E)|/|Q \setminus E|$ — coverage of question tokens not yet present in evidence;
- **doc_len** $= \log(1 + \text{token count})$, normalized;
- **turn_idx**, **n_evidence** — search step and number of consumed documents, normalized.

Semantic variants of alignment/novelty features (E5 embedding based) were additionally used in the gate campaign (Sec. 10). The refit coefficients reported in the main paper (Fig. 3; Table S2) are logistic models without intercept over standardized features within CE-conflicting pairs.

7 Exact-Value Tables Referenced from the Main Paper

Table 2: Table S1. Prospective headroom beyond CE relevance selection (main paper Fig. 2). Oracle–CE is the question-level mean difference in downstream answer F1 with paired-bootstrap 95% CI. Opportunity rates are measured on preregistered failure-enriched stratified samples and are not population-prevalence estimates.

Setting	n (q / states)	ΔF1	95% CI	Opp.
HotpotQA dev (preregistered)	297 / 579	+0.082	[0.062, 0.104]	30.1%
HotpotQA lockbox (single-open)	350 / 686	+0.117	[0.092, 0.142]	29.3%
2WikiMultiHopQA	350 / 819	+0.142	[0.117, 0.168]	26.4%
Qwen2.5-14B trajectories	350 / 838	+0.097	[0.076, 0.120]	32.1%
MuSiQue	349 / 959	+0.097	[0.080, 0.116]	21.8%

Table 3: Table S2. Refit logistic coefficients within CE-conflicting pairs (no intercept, standardized features; main paper Fig. 3). Parentheses give bootstrap sign stability (500 resamples). Query alignment and document length are confirmatory; redundancy and novelty are exploratory.

Setting	Query align.	Doc. length	Redundancy	Novelty
HotpotQA dev	−2.54 (1.00)	−3.37 (1.00)	+1.06 (1.00)	+0.50 (0.96)
Lockbox	−2.45 (1.00)	−2.73 (1.00)	−1.41 (0.998)	−0.21 (0.77)
2Wiki	−1.91 (1.00)	−3.82 (1.00)	+1.14 (0.99)	+1.74 (1.00)
Qwen2.5-14B	−2.93 (1.00)	−2.16 (1.00)	−1.79 (1.00)	−0.29 (0.78)
MuSiQue	−2.19 (1.00)	−2.00 (1.00)	+1.76 (1.00)	+1.18 (1.00)

8 Case Study: Full Trajectories

The main-paper case (Table 1 there) is lockbox state `5ae5d06a...::s1`. *Question:* "From which team did the linebacker for four NFL teams between 1985–1996 get fired by on November 29?" (gold: the Jacksonville Jaguars). *State s_1:* the agent's first search ("linebacker who played for four nfl teams between 1985-1996") returned *Jack Del Rio* ("head coach of the Oakland Raiders ... A linebacker for four NFL teams between 1985–1996"), *Dan Brandenburg*, and *Fredd Young*. The agent's visible reasoning at this point reads: "Jack Del Rio is the only one who fits the criteria ... I need to find out which team fired him on November 29."

Base continuation (no intervention). Four further searches ("Jack Del Rio fired by on November 29", "... fired date", "... fired by which team ...") fail to surface the 2011 Jaguars firing; the agent answers "Oakland Raiders" (F1 0).

CE-selected candidate (pool CE rank 1): *Wilber Marshall* (60 tokens; "... was a linebacker in the National Football League (NFL) for five teams from 1984 until 1995 ..."). Continuation: the agent immediately searches "Wilber Marshall fired on November 29" and answers "Dallas Cowboys" (F1 0). The document is the most query-aligned candidate in the pool but concerns a different entity; it displaces a correct hypothesis.

Useful candidate (pool CE rank 15; *redundancy* family; non-gold): *Kevin Hardy (linebacker)* (70 tokens; "... he played professionally for the Jacksonville Jaguars, Dallas Cowboys, and Cincinnati Bengals of the NFL"). Continuation: the agent keeps the Del Rio hypothesis, issues one search ("Jack Del Rio fired by on November 29"), and answers "Jacksonville Jaguars" (F1 1). Interpretable features: query alignment 0.50 (identical to the CE-selected document's 0.50), novelty 0.51, redundancy 0.35.

Matched control (*Fredd Young*; same BM25 rank bucket, 63 tokens). Continuation reproduces the base failure pattern (four searches, wrong answer), so the useful candidate's effect is not attributable to generic context expansion. (This control happens to be a document already present in the context; see the note on control matching below.)

Caveats. PIU records the realized downstream effect: the agent's visible reasoning never cites the injected Kevin Hardy document, so the mechanism is the availability of the missing entity in context rather than explicit citation. This state is outside the 20% replay-audited subset (single greedy continuation per arm). The case was selected from lockbox states satisfying (i) base and CE-selected outcomes F1 < 0.3, (ii) a non-gold, non-CE-top candidate with F1 ≥ 0.9 and pool CE rank ≥ 4; 25 states met these criteria and this one was chosen for interpretability of the mechanism.

Note on control matching. Matched controls are drawn from the BM25 top-100 under q_t subject to the rank-bucket, length, title-overlap, and non-gold constraints; documents already present in the agent's context are *not* excluded by the matching code. This occurs for 8.4–10.0% of controls across settings (lockbox: 648 of 7,120) and amounts to a near-duplicate injection, i.e., a conservative control. All confirmatory endpoints (opportunity, Oracle–CE headroom, pairwise labels, specialist and gate evaluations) are computed from the raw intervention outcome $Y(s_t, E_{<t} \cup \{d\})$ and are unaffected; controlled PIU is a diagnostic quantity.

9 Additional Figures

Figure 1 summarizes the frozen-specialist transfer together with the coefficient sign matrix across all five settings (the coefficient plot itself appears as main-paper Fig. 3; exact values in Table S2).

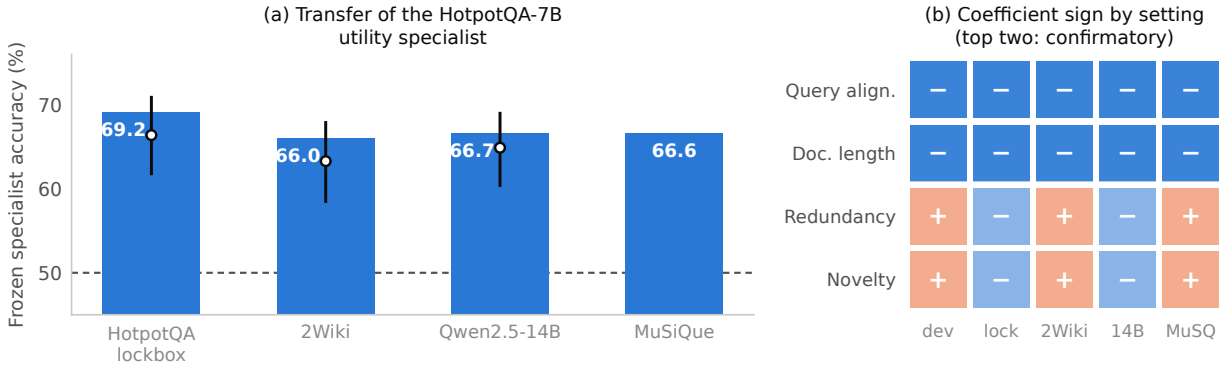


Figure 1: Structure replication. (a) Pair-level transfer of the frozen HotpotQA-7B conflicting-condition specialist with question-clustered 95% confidence intervals where preregistered. (b) Coefficient sign by setting; the top two rows are the confirmatory signatures.

10 Additional Quantitative Results

Effect-size decomposition (full). Table 4 reports opportunity rates at increasing thresholds of the best available downstream improvement.

Table 4: Fraction of evaluated states whose best candidate exceeds the given downstream ΔF1 threshold.

Setting	$\Delta F1 > 0$	≥ 0.05	≥ 0.10	≥ 0.25
HotpotQA lockbox	29.3%	29.3%	28.7%	27.8%
2Wiki	26.4%	26.1%	26.0%	24.8%
Qwen2.5-14B	32.1%	31.9%	31.1%	29.1%

Exchangeability check: CE is a serious baseline. Under a null in which the CE choice is exchangeable with a random pool member, the oracle−random gain on the lockbox would be +0.155 (95% null interval $[0.141, 0.169]$), whereas the observed oracle−CE gain is +0.117. CE selection is therefore significantly better than random selection within the same pools; the reported headroom is measured *beyond* a genuinely informative baseline.

Gate and predictor campaign summary. Deployment-time models were developed strictly on development data and frozen before lockbox evaluation. Standalone pairwise utility prediction reached 57.0% (v1) and at most 59.3% (v2) accuracy; on CE-disagreement pairs it remained below chance (39–47%), while CE relevance alone reached 53.1% and CE combined with measured-utility features 62.1%. For explicit CE-conflict identification, four gate families of increasing capacity (lexical/retrieval statistics; CE-score distribution and probe signals; semantic state–candidate representations; a fine-tuned transformer gate) converged to development AUC 0.61–0.69; the frozen ensemble achieved 0.641 on the lockbox, and the resulting deployable mixture recovered approximately none of the oracle-condition routing advantage (recovery ≈ -0.02; main paper Fig. 4). Oracle-condition routing itself is stable at 71–76% across all five settings (76.0 / 76.2 / 71.2 / 75.2 / 71.7%).

Gold is not sufficient either. Among gold-labeled candidate documents in the main Gate-1 analysis, 74.2% did not improve the downstream outcome at their state (62.9% in the preregistered Gate-2 sample); in a weakened-agent stress condition this fell to 37.5%, showing that even ground-truth supporting evidence has state-dependent utility.

Context pollution. Omission interventions (I1) show that removing already-consumed non-gold evidence improves downstream answer F1 by +0.057 on average ($n = 330$ omission arms), while removing gold evidence harms performance more than removing non-gold evidence — an independent sanity check of the measurement and complementary evidence that additional relevant-looking context is not uniformly beneficial.

Search-termination utility. Interventions can change search behavior itself: the fraction of continuations that terminate search immediately after the intervention rises from 7.1% to 12.9% under the stress condition, indicating that part of a document's value lies in enabling the agent to stop searching.

Where opportunities live. Opportunities concentrate in base-failure states (37.5% of failure states vs. 0.8% of success states in the development analysis), motivating the failure-enriched confirmatory sampling; the representative (unenriched) development sample retains an 18.5% opportunity rate.

Cost accounting. 96.2% of rollout token cost is prompt tokens (context replay), implying that amortized approximations of prospective intervention should target context reuse rather than generation length.

11 Reproducibility

Software and hardware. vLLM 0.10.2, PyTorch 2.8.0 (CUDA 12.8), Transformers 4.55.2, `bm25s` 0.3.10, scikit-learn 1.9.0, Python 3.11; NVIDIA H100 80GB. Greedy decoding throughout; fixed seeds for sampling, stratification, and control matching.

Scale of the study. Per-setting confirmatory sizes (questions / states / conflicting pairs): HotpotQA dev 297 / 579 / —; lockbox 350 / 686 / 1,639; 2Wiki 350 / 819 / 1,684; Qwen2.5-14B 350 / 838 / 1,930; MuSiQue 349 / 959 / 2,014. The cross-fitted oracle uses 150 fresh questions with two independent rollout sets.

Code and records. Anonymized code, the preregistration document, and frozen per-gate reports are available at the repository linked in the abstract. All confirmatory numbers in the main paper trace to a preregistered criterion and a frozen report file in that repository.